



UNIVERSITY OF JAFFNA

FACULTY OF ENGINEERING

END SEMESTER EXAMINATION – SEPT 2024 – SEMESTER VI

EC9630 – MACHINE LEARNING

Date: 10/09/2024

Duration: 2 hours

- I. There are 04 questions in five pages and answer all questions.*
- II. The marks for each question are given in parentheses.*
- III. Scientific electronic calculators are permitted.*
- IV. Mention the assumptions.*

QUESTIONS

1. [Total – 25 Marks]
 - 1.1. Sri Lanka aims to achieve a 70% renewable energy target by 2030. To support this goal, we are currently installing rooftop solar systems to decrease reliance on conventional power sources. However, at night, we still depend on conventional power. We have observed that power consumption on the main grid is low between 1 a.m. and 4 a.m.
 - a. To support the country's renewable energy goals, you are asked to develop a prediction system for estimating the daily energy generation of rooftop solar systems installed in the Faculty of Engineering. You have a dataset that includes features such as timestamps, installed capacity, panel efficiency, inverter efficiency, solar irradiance, thermal resistance, and more. **Briefly explain how you would approach this problem using a linear regression algorithm** . [6 Marks]
 - b. You can get real-time data from the weather station and the roof-top solar system. How can you improvise your prediction system using this real-time data? [3 Marks]
 - c. Briefly explain, using plots and equations, how the linear regression algorithm works with multiple independent variables to predict a single dependent variable. [4 Marks]

1.2. You are requested to build a machine that scans the stomach (uterus) of a female to identify the pregnancy by themselves. Assume you are planning to implement the system using ultrasonic scanners. Before implementing a physical system, you worked with sample data collected from a reputed medical repository.

a. State two non-parametric machine learning algorithm. [3 Marks]

b. Briefly explain how you can apply logistic regression algorithm for this image dataset. [6 Marks]

c. After training your model, you test it on a new dataset of 500 images and achieve 80% classification accuracy. Using the provided confusion matrix, critically analyse why this model may not be suitable for your specific application. [3 Marks]

		True Class	
		Pregnant	Not Pregnant
Predicted Class	Pregnant	70	5
	Not Pregnant	95	330

2. [Total – 25 Marks]

2.1. Assume you have data from students who studied in the Faculty of Engineering at the University of Jaffna in the year from 2014 to 2018. A set of features (A/L results, semester results, laboratory marks, etc.) extracted from each student's academic records. Our goal is to decide the suitable department for new students.

a. You decide to use a neural network to solve this problem.

You have two choices:

A. Train separate neural networks for each of the departments

B. Train a single neural network with one output neuron for each department with a shared hidden layer.

Which method do you prefer? Justify your answer. [3 Marks]

b. If you use a neural network with a single output neuron for each department and a shared hidden layer, how output layer classify students into suitable departments? [3 Marks]

- c. You rank all four specializations for each student. How do you modify the output layer of the neural network from 2.1.b to achieve this? [3 Marks]
- d. Faculty administration decided to allow the students who show the outstanding performance in their first 3 semester to their preferred department. Outstanding students are the top 25% of the student population. Which machine learning algorithm would you combine with this neural network to achieve best performance? Justify your answer. [4 Marks]

2.2.

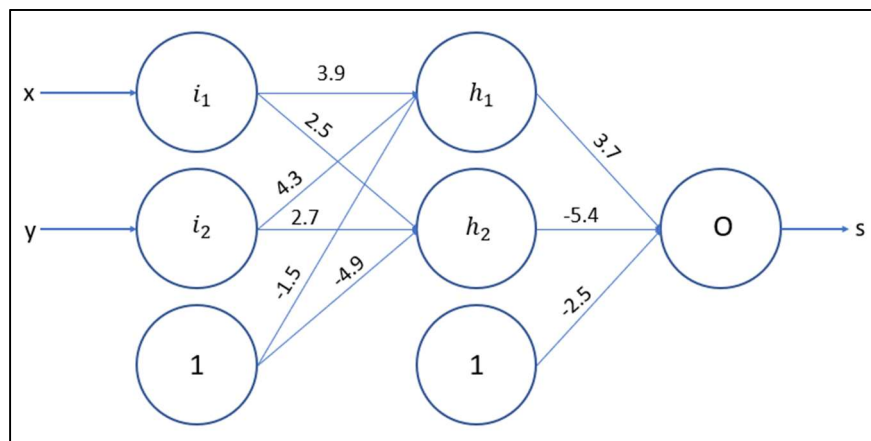


Figure 1: Simple neural network

- a. Prove that neural network mentioned in figure 1 is linear. [3 Marks]
- b. Calculate the output (S) of neuron (O) if input $x = 1.0$ and $y = 0$ with linear combination of weights. [3 Marks]
- c. Briefly explain how to introduce nonlinearity into this neural network, using equations to illustrate the process [3 Marks]
- d. Calculate the output of the network after applying nonlinearity. Consider input $x = 1.0$ and $y = 0$. [3 Marks]

3.

[Total – 25 Marks]

3.1.

- a. Briefly explain how support vector machine works. [3 Marks]
- b. You have clearly separable positive label and negative label data. **State** what will be the **shape** of the hyper plane if you classify those points in R^2 (X, Y) plane and R^3 (X, Y, Z) plane? Justify your answer. [3 Marks]
- c. You have a dataset with both positively and negatively labelled data points that are not separable in the R^2 plane. Briefly explain how SVM can be used to classify this dataset. [4 marks]

3.2. You are given two separate set of data points in R^2 .

$$\text{Positive Label} = \left\{ \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 4 \\ -6 \end{pmatrix}, \begin{pmatrix} 6 \\ 3 \end{pmatrix}, \begin{pmatrix} 6 \\ -3 \end{pmatrix}, \begin{pmatrix} 5 \\ 1 \end{pmatrix} \right\}$$

$$\text{Negative Label} = \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \end{pmatrix} \right\}$$

- a) Plot the data points in a 2-D plan. [2 Marks]
- b) Identify the coordinates of Support vectors (S_i) and State the augmented vectors $\tilde{S}_i$ with bias = 1. [2 Marks]
- c) For the augmented vectors in the Question 3.1.b, write the equations for calculating alpha (α_i) for each support vector. [2 Marks]
- d) Calculate $\tilde{w}$ vector. ($\tilde{w} = \sum_{i=1}^n \alpha_i \tilde{S}_i$) [5 Marks]
- e) Draw the hyper plane and mark the margins with the equation. [3 Marks]
- f) Choose a support vector which can be removed from the model without making any changes in the hyperplane. [1 Marks]

4.

[Total – 25 Marks]

4.1.

a. Compare and contrast Partition (Flat) clustering algorithms and Hierarchical clustering algorithms. [3 Marks]

b. With a help of a diagram briefly explain how K-Means algorithm works? [6 Marks]

c. "When performing K-means clustering, if a cluster center (centroid) ends up at an outlier point, it can significantly affect the clustering results". Critically argue this statement. [6 Marks]

4.2. You have a data set with **many features**. When you consider X and Y features you have the following plot. You must analyze the plot and answer the following question.

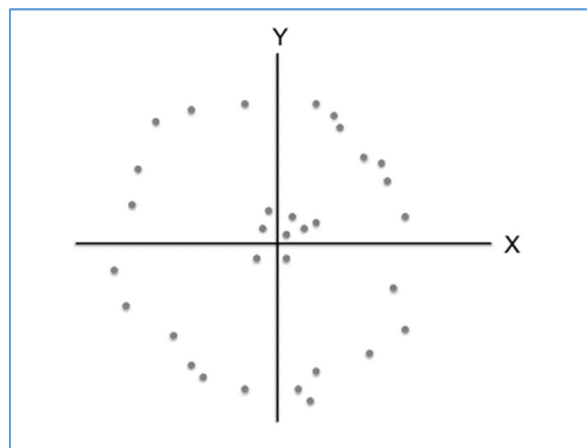


Figure 2: X vs Y feature plot

a. What can be the K value for this problem? [1 Marks]

b. Can you accurately cluster this dataset using k-Mean algorithm? Justify your answer [6 Marks]

c. State 2 approaches for clustering data plotted in figure 2. [3 Marks]

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